

OC IT Projects — Chapter 6

Virtualization & Cloud Computing — Quick Review + Quiz

Based on the Chapter 6 training content we studied.

1. Quick Review

Virtual Network Devices — Network devices can be implemented virtually instead of relying only on physical hardware.

Virtual Servers — A server can be implemented as a Virtual Machine, allowing multiple independent virtual server environments to run on physical hardware.

Hypervisor — Technology used to create/manage Virtual Machines on physical hardware. Examples listed: VMware ESXi, Microsoft Hyper-V, VirtualBox, VMware Workstation.

VoIP — Voice over IP: voice calls are carried over an IP network.

SIP — Session Initiation Protocol; associated with call request/session setup.

RTP — Real-time Transport Protocol; associated with real-time voice/media after the call is established.

PBX — Private Branch Exchange, presented in the training in the context of VoIP and phone systems.

Cloud Computing — On-demand network access to shared computing resources.

Private Cloud — Systems/users access resources inside the private cloud/system.

Public Cloud — Cloud resources are accessed through public networks such as the Internet.

Hybrid Cloud — Combination of Private Cloud and Public Cloud.

NaaS — Network as a Service. Examples: Virtual Routers, Virtual Switches, Virtual Firewalls.

IaaS — Infrastructure as a Service; associated with System / IT Administrators.

PaaS — Platform as a Service; associated with Developers who create/control applications.

SaaS — Software as a Service; associated with End Users.

2. Must-Memorize Map

- Hypervisor → Virtual Machines
- SIP → Call setup / signaling
- RTP → Real-time voice/media
- Private + Public → Hybrid Cloud
- NaaS → Network → Virtual Routers / Switches / Firewalls
- IaaS → Infrastructure → IT Administrators
- PaaS → Platform → Developers
- SaaS → Software → End Users

3. Chapter 6 Quiz — 20 Questions

Use for later revision; answer before checking the key.

Q1. What is the main purpose of a Hypervisor?

- A) To route packets between networks
- B) To provide Internet access
- C) To create/manage Virtual Machines on physical hardware
- D) To encrypt VoIP calls

Q2. Which is listed as an example of a Hypervisor/virtualization platform?

- A) VMware ESXi
- B) Cisco IOS
- C) Apache HTTP Server
- D) MySQL

Q3. What is a Virtual Server?

- A) A physical server with multiple network cards
- B) A server implemented as a virtual machine rather than requiring separate physical hardware
- C) A router configured as a server
- D) A cloud storage account

Q4. Which protocol is used for the VoIP call request/session setup?

- A) RTP
- B) SIP
- C) HTTP
- D) FTP

Q5. After the VoIP call is established, which protocol is associated with real-time media?

- A) SIP
- B) DNS
- C) RTP
- D) DHCP

Q6. One physical server must run several independent server environments. What is the key enabler?

- A) PBX
- B) Hypervisor
- C) SIP
- D) Public Cloud

Q7. Which is a Cloud deployment model?

- A) Hybrid Cloud
- B) Virtual Cloud
- C) Application Cloud
- D) Network Cloud

Q8. What characterizes a Private Cloud in the training?

- A) Accessible only through public Internet
- B) Systems/users access resources inside the private cloud/system
- C) Must combine two public clouds
- D) Available only to developers

Q9. Private cloud combined with public-cloud resources is:

- A) Public

- B) Private
- C) Hybrid
- D) NaaS

Q10. Which model provides Network functionality as a Service?

- A) IaaS
- B) SaaS
- C) PaaS
- D) NaaS

Q11. Virtual Routers, Virtual Switches, and Virtual Firewalls are examples of:

- A) NaaS
- B) SaaS
- C) PaaS
- D) IaaS

Q12. Which Cloud Service Model is associated with System / IT Administrators?

- A) SaaS
- B) IaaS
- C) PaaS
- D) NaaS

Q13. Which Cloud Service Model is primarily associated with Developers?

- A) IaaS
- B) SaaS
- C) PaaS
- D) NaaS

Q14. Which Cloud Service Model is associated with End Users?

- A) PaaS
- B) NaaS
- C) IaaS
- D) SaaS

Q15. A developer wants to create/control applications without managing the entire underlying infrastructure. Which model fits?

- A) PaaS
- B) IaaS
- C) NaaS
- D) SaaS

Q16. Which statement best describes Cloud Computing according to the training?

- A) Running applications only locally
- B) On-demand network access to shared computing resources
- C) Using only physical servers
- D) Connecting two LANs with a router

Q17. Which combination is correct?

- A) SIP → real-time media, RTP → call setup
- B) SIP → call setup, RTP → real-time media
- C) SIP → cloud storage, RTP → virtualization
- D) SIP → routing, RTP → DNS

Q18. Employees use a ready-made cloud software application without managing infrastructure or developing it. Which model fits?

- A) IaaS
- B) PaaS
- C) NaaS
- D) SaaS

Q19. IT administrators need control over cloud infrastructure rather than finished software. Which model fits?

- A) SaaS
- B) PaaS
- C) IaaS
- D) NaaS

Q20. A company has a private cloud but uses public-cloud resources during peak traffic. Which description is accurate?

- A) Public Cloud + SaaS
- B) Private Cloud + IaaS
- C) Hybrid Cloud
- D) NaaS + PaaS

4. Answer Key

1-C | 2-A | 3-B | 4-B | 5-C

6-B | 7-A | 8-B | 9-C | 10-D

11-A | 12-B | 13-C | 14-D | 15-A

16-B | 17-B | 18-D | 19-C | 20-C

5. Final Revision Card

- Virtualization → multiple virtual environments can run on physical hardware.
- Hypervisor → creates/manages Virtual Machines.
- SIP = call setup/signaling; RTP = real-time media.
- Private + Public = Hybrid Cloud.
- NaaS = Network as a Service.
- IaaS = Infrastructure as a Service → IT Administrators.
- PaaS = Platform as a Service → Developers.
- SaaS = Software as a Service → End Users.
- Session result: 20/20 (100%).